

Single-Drive Flat-Top Near-Visible Electro-Optic Frequency Comb Generation in Lithium Tantalate

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Abstract We demonstrate a single-drive flat-top electro-optic comb generator at 798 nm in thin-film lithium tantalate. The device generates 49 comb lines, with a 5 dB bandwidth comprising 25 lines at an 18 GHz microwave drive. ©2026 The Author(s)

Instruction

Optical frequency combs (OFCs) are light sources that generate multiple, discrete, coherent, and evenly spaced frequency lines. Chip-scale systems of optical frequency combs, enabled by recent advancements in photonic integrated circuits, offer significant advantages in compactness and cost compared to bulk optics systems. Over the past decades, the applications of on-chip OFCs have expanded to encompass various fields, including optical communications [1], optical computing [2], neural networks [3], light detection and ranging (LiDAR) [4], and spectroscopy [5].

Currently, on-chip frequency comb generation is dominated by mode-locked lasers, Kerr combs, and electro-optic (EO) combs. Among these, EO combs offer flexible control of comb spacing and central wavelength. However, conventional implementations using cascaded phase and amplitude modulators to achieve a flat spectrum often require external phase synchronization of multiple microwave sources, leading to increased system complexity and power consumption. Furthermore, thin-film lithium niobate, the dominant platform, suffers from mode hybridization at short wavelengths, resulting in high bending loss and mode hybridization.

Here, we present thin-film lithium tantalate (TFLT) as a superior platform for EO comb generation. TFLT offers EO performance comparable to lithium niobate while providing improved optical properties, including a wider bandgap and negligible birefringence [6,7]. By combining TFLT with a domain inversion technique, we demonstrate a single-drive flat-top EO comb generation at 798 nm, which simplifies the system architecture by eliminating the need for complex microwave control. The device is compact and easy to operate, with an 8-mm modulation length and an ultralow optical loss of approximately 0.3 dB. It generates 49 comb lines, including 25 lines within a 1-nm 5 dB bandwidth under an 18 GHz and 35 dBm microwave drive. The comb spacing is

tunable from 18 to 40 GHz, with the upper limit set by the microwave amplifier.

Principle and Design

Figure 1a illustrates the schematic of the TFLT single-drive flat-top EO comb generator, configured as a Mach-Zehnder modulator (MZM) with a domain inversion segment in the lower arm. The optical signal is equally divided into two interferometer arms by a multimode interferometer (MMI), with subsequent recombination through a symmetric output MMI.

The optical field output of the proposed single-drive flat-top EO comb based on domain inversion waveguides is as follows:

$$A(t) = \frac{A_0}{2} e^{i\omega_c t} \left(e^{iK(V_1(t)+V_{b,1})} + e^{iK(V_2(t)+V_{b,2})} \right) \quad (1)$$

where $A(t)$ is the output optical field, A_0 is the optical amplitude, ω_c is the optical carrier frequency, K is the modulation index ($K = \pi/V_\pi$). The modulated electric field $V_1(t) = V_0 \sin(\omega_m t)$, V_0 is the peak voltage, ω_m is the modulation frequency. The lower MZM arm incorporates a poled region of length d to realize opposite phase modulation (Figure 1b). The poling duty cycle $p = d/L$, where L represents the total modulation arm length. $V_2(t) = (2p-1)V_0 \sin(\omega_m t)$. $V_{b,1}$ is the bias voltage applied on the upper arm, $V_{b,2}$ is the bias voltage applied on the lower arm, $V_{b,2} = (2p-1)V_{b,1}$. The frequency-domain expression of the modulated output optical field (1) is given by:

$$A(\omega) = \frac{A_0}{2} \sum_{n=-\infty}^{\infty} [J_n(KV_0) e^{iKV_{b,1}} \delta(\omega - n\omega_m - \omega_c) + J_n((2p-1)KV_0) e^{i(2p-1)KV_{b,1}} \delta(\omega - n\omega_m - \omega_c)] \quad (2)$$

The flat-top EO comb achieves maximal pump-to-comb conversion efficiency when the modulation indices between the two MZM arms simultaneously satisfy the condition $KV_0 - |(2p-1)KV_0| = \frac{\pi}{2}$, $KV_{b,1} - |(2p-1)KV_{b,1}| = \frac{\pi}{2}$ [8]. Normally, the difference in modulation indices between the two arms is adjusted using discrete microwave sources with different powers, which

is complicated and difficult to control. In contrast, we achieve the flatness condition by engineering the poling duty cycle.

By optimizing the design through finite-difference time-domain simulations (Fig. 1c), the MMI demonstrates spectral uniformity and low insertion loss over a broad wavelength range from 700 nm to 900 nm. To support high-speed modulation under microwave drive, a coplanar traveling-wave electrode configuration was designed. Figure 1d presents the cross-section of the electro-optic modulator, where optical waveguides traverse through the gaps of a ground-signal-ground (GSG) coplanar waveguide. Devices were fabricated on a 400-nm x-cut TFLT on a 3- μm silica and a 525- μm -thick Si handle substrate, with an etch depth of 200 nm. To maintain high modulation efficiency at high RF frequencies, the parameters of the traveling wave electrode and waveguides were optimized. Waveguide width w_g of 1 μm , electrode thickness of 600 nm, signal electrode width w_s of 14 μm , ground electrode width w_G of 50 μm , and electrode gap of 3.5 μm were used after careful optimization, achieving microwave and optical group refractive index matching of ~ 2.3 , traveling wave electrode impedance of $\sim 45 \Omega$.

The simulated EO bandwidth, as shown in Figure 2a, achieves a 4 dB bandwidth exceeding 60 GHz, which indicates that the V_π would not drop significantly at high RF frequencies. To accurately determine the modulation index, we characterized the static EO response using a 1-kHz triangular-wave voltage sweep. The

measured results plotted in Figure 2b indicate that the half-wave voltage V_π for the 8-mm EO modulator is 1.14 V. Numerical simulations identified an optimal duty cycle of $p = 10\%$ for achieving the flattest comb spectrum, as shown in Figure 2c. The corresponding spectrum features a total of 53 comb lines, with 25 lines contained within the 5 dB bandwidth.

Results

As verified by piezo-response force microscopy (PFM) phase imaging (Figure 3a), domain inversion was successfully achieved in the TFLT waveguides through the use of 400 V poling voltage pulses. The experimental setup for measuring the fabricated EO comb chip is shown in Figure 3b. The pump wave was generated by a high-power continuous-wave (CW) fiber laser source with a fixed wavelength of 798 nm, and the input polarization was aligned to the TE mode using a fiber polarization controller. The fiber-waveguide coupling loss is 8 dB per facet at 798 nm. The single microwave source and amplifier delivered the RF signal to the MZM through a GSG probe, terminated with an off-chip 50 Ω load (Figure 3c).

As shown in Figure 3d, a total of 49 comb lines spaced at 18 GHz were generated under a microwave drive power of 35 dBm and a modulation length L of 0.8 cm. The number of comb lines is limited by the output power of our microwave amplifier. Future implementations could achieve a higher modulation efficiency by increasing the device length or adopting capacitively loaded

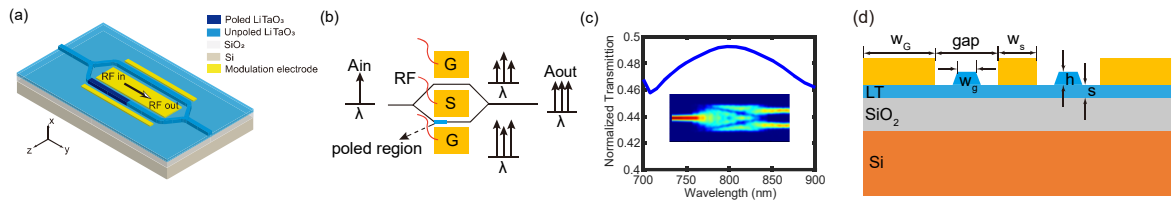


Fig. 1: (a) A schematic illustration of the single-drive flat-top EO comb. Domain polarization inversion in a section of the lower arm. (b) Working principle of flat-top optical frequency comb generation using a single-drive MZM. (c) The transmission spectrum of the 1×2 MMI from 700 to 900 nm, demonstrating less than 0.1 dB insertion loss. The inset shows the simulated optical field propagation. (d) A schematic cross-sectional view of the proposed EO modulator.

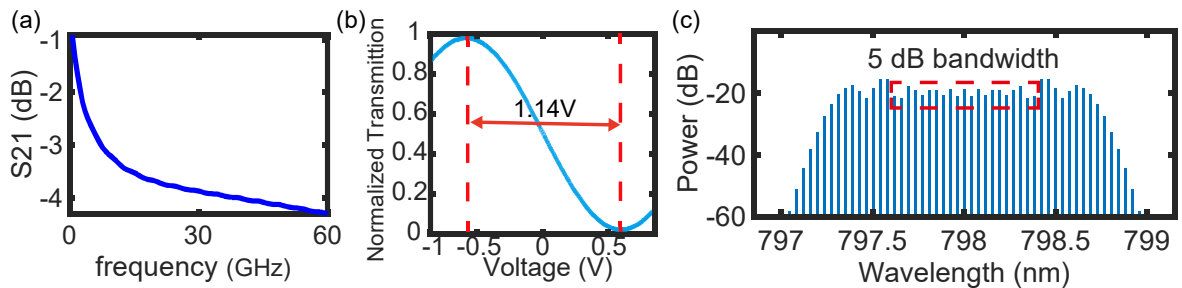


Fig. 2. (a) Simulated EO bandwidth. (b) Measured optical transmission as a function of driving voltage. (c) Simulated EO comb spectra for 10% poling duty cycles.

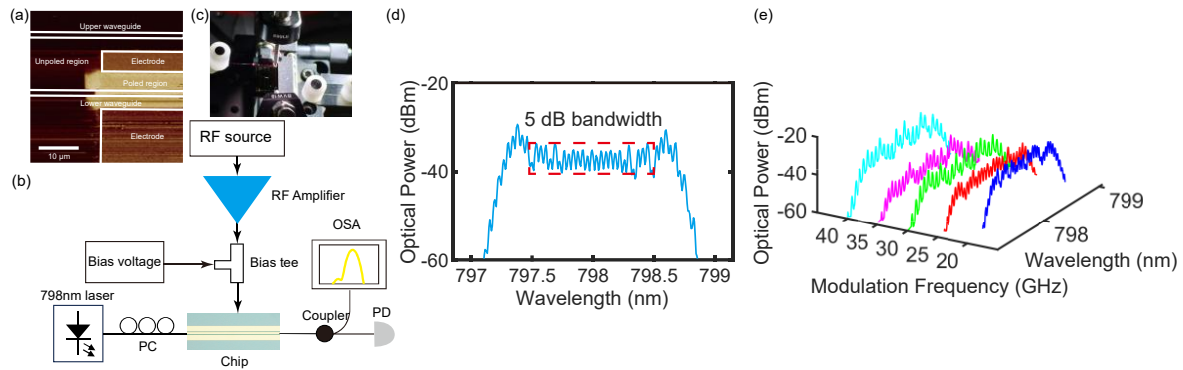


Fig. 3. (a) PFM phase image of a TFLT waveguide after poling. Domain inversion is observed at the surface of the lower waveguide. (b) Experimental setup. (c) Image of the experimental setup. (d) Optical transmission spectra of the flat-top EO comb under 10% domain inversion duty cycle. (e) Measured EO comb spectra at different RF driving frequencies of 20, 25, 30, 35, and 40 GHz.

traveling-wave electrodes, reducing microwave propagation losses. The resulting spectrum exhibits excellent symmetry and flatness, with 25 comb lines spanning 1 nm within a 5 dB optical bandwidth, which is consistent with the simulation in Figure 2c. Moreover, the present device can efficiently be driven with a broad range of microwave frequencies as well. As shown in Figure 3e, microwave frequencies from 20 to 40 GHz are demonstrated at a microwave power of 31 dBm. All measured OFC spectra feature more than 40 comb lines and exhibit flat-top spectra.

Conclusions

In this work, we present a novel design strategy for generating a flat-top EO comb on the TFLT platform using only a single microwave drive and a single modulator. The proposed device exhibits a broad flat-top EO comb with a low insertion loss of approximately 0.3 dB and a widely tunable comb spacing ranging from 18 to 40 GHz. The device generates up to 49 comb lines at a repetition rate of 18 GHz, with a 5 dB optical bandwidth of 1 nm encompassing 25 comb lines. Owing to its simple architecture and robust operation, the proposed EO comb provides a promising solution for applications in high-speed optical communications, microwave photonics, LiDAR, sensing, and spectroscopy.

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